

**SUNJOY AGRO**

NINGBO SUNJOY AGROSCIENCE CO., LTD.

Add: No.1165 Beihai Road, Chemical Industry Zone of Ningbo,
Zhenhai District, Ningbo, Zhejiang Province, China
Tel: 86-574-87774749 87760080
Fax: 86-574-87774751

MATERIAL SAFETY DATA SHEET

SECTION 1. PRODUCTION IDENTIFICATION

Product Name: Tebuconazole 430 g/L SC
Company Name: Ningbo Sunjoy Agrosience Co., Ltd.
Active Ingredient: Tebuconazole
Chemical Name: (RS)-1-*p*-chlorophenyl-4,4-dimethyl-3-(1H-1,2,4-triazol-1-ylmethyl)pentan-3-ol (IUPAC)
Chemical Formula: C₁₆H₂₂ClN₃O
CAS #: 107534-96-3
Chemical Class: Triazole
Usage: Fungicide
Formulation: Suspension Concentrate

SECTION 2. COMPOSITION/INFORMATION ON INGREDIENTS

Component	CAS Number	Concentration (w/v)
Tebuconazole	107534-96-3	430 g/L
Other inert ingredients	/	Fill to 1 L

SECTION 3. HAZARDS IDENTIFICATION

Harmful if swallowed. Irritating to skin. Risk of damage to eyes. Non-flammable.

SECTION 4. FIRST AID MEASURES

If inhaled, remove from exposure; rest and keep warm. Oxygen or artificial respiration if needed. Seek medical attention.

If skin contact, Remove contaminated clothing immediately and wash affected areas with soap and plenty of water. Seek medical attention.

If eye contact, immediately irrigate thoroughly with water, including under eyelids.



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Seek medical attention immediately.

If ingested, wash out mouth with water. Do not induce vomiting. Seek medical attention.

Immediate treatment/ antidote: Treatment supportive and symptomatic.

Delayed effects: No delayed effects likely

SECTION 5. FIRE FIGHTING MEASURES

Specific hazards: Do not contaminate watercourses.

Suitable extinguishers: Water spray, dry chemical CO₂ or dry powder as suitable for fire conditions.

Unsuitable extinguishers: High volume water jet (contamination risk)

Personal protective equipment: Wear self-contained breathing apparatus. See section 8

Hazardous combustion products: Hydrogen sulphide, oxides of carbon, sulphur and nitrogen.

Special equipment or procedures for fire fighting: None

SECTION 6. ACCIDENTAL RELEASE MEASURES

Safety precautions:

Wear suitable personal protective equipment - see section 8.

Environmental precautions:

Avoid contamination of soil, or entry into drains and watercourses. Inform local authorities if the material enters drains, rivers or sewers, or if soil or vegetation is contaminated in a public place.



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Clean-up procedure:

After spillage or leakage: Clean up straight away by scoop or vacuum. Do not flush with water. Absorb spillages on to sawdust, peat or other inert material. Shovel into suitable closable container for disposal. Use personal protective equipment. Clean floors and contaminated tools with a damp cloth and place any used cleaning materials into closable containers for disposal.

SECTION 7. HANDLING AND STORAGE

Industrial hygiene:

Wash hands, face and exposed skin before breaks and immediately after handling the product. Avoid contact with skin or eyes.

Handling:

Protection against fire and explosion: Keep away from heat and sources of ignition.

Storage:

Keep containers tightly closed in a dry, cool and well-ventilated place inaccessible to unauthorized persons.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Exposure limits: N/A

Monitoring method: N/A

Personal protective equipment:

Respiration: full respirator with ABEK-P2 filter;

Eyes: goggles;

Hands: nitrile gauntlet gloves minimum 0.5mm thick;

Other: overalls or protective apron.

Hygiene measures:

Wash hands and exposed skin before breaks and immediately after handling the product.



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SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance:	Off-white suspension liquid
Odor:	Chalky
Boling point:	Not available
Specific gravity or density:	1.12 at 20°C
Flash point:	> 100°C
pH:	5.0 – 8.0
Corrosivity:	Not corrosive
Explosivity:	Not explosive
Stability:	Stable at least two years under normal storage conditions.

SECTION 10. STABILITY AND REACTIVITY

Stability:	Stable under ambient storage conditions
Conditions to Avoid:	Extremes of temperature up to >100°C.

Incompatibility Materials:

This product is unstable in acids or bases.

Hazardous decomposition product:

Hydrogen sulphide, oxides of carbon, sulphur and nitrogen

Danger of polymerization: None known

SECTION 11. TOXICOLOGICAL INFORMATION

Product is of low toxicity to mammals

Acute oral LD ₅₀ ,	rat > 2000 mg/kg;
Acute dermal LD ₅₀ ,	rat > 2000 mg/kg;
Acute inhalation LC ₅₀ ,	rat > 0.70 mg/l;
Skin irritation (rabbit),	rabbit, not an irritant;



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Eye irritation (rabbit), rabbit, not an irritant;
Sensitization, guinea pig, not a skin sensitizer

SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity data for toxicity of Tebuconazole technical:

Birds: Acute oral LD50 for male Japanese quail 4438, female Japanese quail 2912, bobwhite quail 1988 mg/kg b.w. Dietary LC50 (5 d) for mallard ducks >4816, bobwhite quail >5000 mg/kg feed.

Fish: LC50 (96 h, flow-through) for rainbow trout 4.4, bluegill sunfish 5.7 mg/l.

Daphnia: LC50 (48 h, flow-through) 2.79 mg/l.

Algae: ErC50 (72 h, static) for *Pseudokirchneriella subcapitata* 3.80 mg/l.

Other aquatic spp.: EC15 (28 d) for *Chironomus riparius* 2.51 mg/l.

Bees: LD50 (48 h, oral) >83 µg/bee; (contact) >200 µg/bee.

Worms: Acute LC50 (14 d) for *Eisenia foetida* 1381 mg/kg dry soil.

Other beneficial spp.: No adverse effects on ground dwellers, e.g. ground beetles (*Poecilus cupreus*) (adults and larvae) or on foliage dwellers, e.g. ladybirds (*Coccinella septempunctata*), up to 375 g/ha (EW 250 formulation).

SECTION 13. DISPOSAL CONSIDERATIONS

Substance:

Via authorized waste disposal contractor to a suitable approved waste disposal site or incinerator. Do not contaminate ponds, waterways or ditches with chemical or used container

Container: As substance.

SECTION 14. TRANSPORT INFORMATION

Land transport ADR/RID

UN number: 3082

Proper Shipping Name: Environmental hazardous substance, liquid, n.o.s. (contains Tebuconazole)



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Class: 9
Subsidiary Risk: Not regulated
Packaging group: III

Marine transport IMDG

UN number: 3082
Proper Shipping Name: Environmental hazardous substance, liquid, n.o.s. (contains Tebuconazole)
Class: 9
Subsidiary Risk: Not regulated
Packaging group: III
Marine pollutant: Yes

Air transport ICAO/IATA

UN number: 3082
Proper Shipping Name: Environmental hazardous substance, liquid, n.o.s. (contains Tebuconazole)
Class: 9
Subsidiary Risk: Not regulated
Packaging group: III

SECTION 15. REGULATOR INFORMATION

Risk phrase(s): Harmful if swallowed
Safety phrases: Keep out of reach of children. When using, do not eat, drink or smoke. Keep away from food, drink and animal feeding stuffs. Avoid contact with skin. Wear suitable protective clothing, face protection (face shield) and gloves.

SECTION 16. OTHER INFORMATION

Revise date: Jan. 27, 2015
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